

Si-Woo Eum

Ph.D. Candidate • CryptoCraft Lab • Hansung University

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Profile

Ph.D. candidate in Information and Computer Engineering at Hansung University, specializing in platform-optimized cryptographic implementation. Active researcher in post-quantum cryptography (PQC) benchmarking and embedded/GPU-accelerated cryptographic engineering, with publications across SCI(E) journals and international conferences, and contributions to multiple national cryptography research projects. Recent work also explores AI-agent-assisted workflows for cryptographic implementation, benchmarking, and test-harness generation.

Research Interests

- Platform-optimized cryptographic implementation
- Post-quantum cryptography (PQC) implementation and benchmarking (KpqC, NIST PQC)
- Embedded cryptography on ARM (Cortex-M, ARMv8/NEON) and RISC-V
- GPU/CUDA-accelerated cryptography
- Symmetric-key cryptography and SIMD parallelization
- AI-agent-assisted cryptographic engineering workflows

Education

Mar. 2023 – Present	Ph.D. Candidate in Information and Computer Engineering, Hansung University, South Korea. Advisor: Prof. Hwa-Jeong Seo, CryptoCraft Lab.
Feb. 2023	M.S. in IT Convergence Engineering, Hansung University, South Korea.
Feb. 2021	B.S. in IT Convergence Engineering, Hansung University, South Korea.

Publications

SUMMARY SCI(E) 9 KCI 8 Preprints 3

Research Papers Published in SCI(E)

1. *Accelerating FAEST Signatures on ARM: NEON SIMD AES and Parallel VOLE Optimization.*
Seung-Won Lee, Ha-Gyeong Kim, Min-Ho Song, **Si-Woo Eum**, Hwa-Jeong Seo
Applied Sciences 16(8), 3782, 2026.
2. *Integrating and Benchmarking KpqC in TLS/X.509.*
Min-Joo Sim, Gyeong-Ju Song, **Si-Woo Eum**, Min-Woo Lee, Sung-Yoon Yoon, Anubhab Baksi, Hwa-Jeong Seo
Electronics 14(18), 3717, 2025.
3. *Efficient GPU Parallel Implementation and Optimization of ARIA for Counter and Exhaustive Key-Search Modes.*
Si-Woo Eum, Min-Ho Song, Sang-Won Kim, Hwa-Jeong Seo
Electronics 14(10), 2021, 2025.
4. *Optimizing Hawk Signature Scheme Performance on ARMv8.*
Si-Woo Eum, Min-Woo Lee, Hwa-Jeong Seo
Applied Sciences 14(19), 8647, 2024.
5. *Optimized Quantum Circuit for Quantum Security Strength Analysis of Argon2.*
Gyeong-Ju Song, **Si-Woo Eum**, Hyeok-Dong Kwon, Min-Joo Sim, Min-Woo Lee, Hwa-Jeong Seo
Electronics 12(21), 4485, 2023.
6. *K-XMSS and K-SPHINCS+: Enhancing Security in Next-Generation Mobile Communication and Internet Systems with Hash Based Signatures Using Korean KSC.*
Min-Joo Sim, **Si-Woo Eum**, Gyeong-Ju Song, Yu-Jin Yang, Wai-Kong Kim, Hwa-Jeong Seo
Sensors 23(17), 7558, 2023.
7. *Optimized Implementation of ARGON2 Utilizing the Graphics Processing Unit.*
Si-Woo Eum, Hyun-Jun Kim, Min-Ho Song, Hwa-Jeong Seo
Applied Sciences 13(16), 9295, 2023.
8. *Parallel Implementations of ARIA on ARM Processors and Graphics Processing Unit.*

Si-Woo Eum, Hyun-Jun Kim, Hyeok-Dong Kwon, Min-Joo Sim, Gyeong-Ju Song, Hwa-Jeong Seo
Applied Sciences 12(23), 12246, 2022.

9. *Optimized Implementation of Simpira on Microcontrollers for Secure Massive Learning.*
Min-Joo Sim, **Si-Woo Eum**, Hyeok-Dong Kwon, Kyung-Bae Jang, Hyun-Jun Kim, Hyun-Ji Kim, Gyeong-Ju Song, Wai-Kong Lee, Hwa-Jeong Seo
Symmetry 14(11), 2377, 2022.

Research Papers Published in KCI

1. *Optimal Implementation of PIPO Block Cipher CTR Mode on GPU.*
Min-Ho Song, **Si-Woo Eum**, Sang-Won Kim, Hwa-Jeong Seo
Smart Media Journal (SMJ), 2025.
2. *GPU-Accelerated Exhaustive Search and Counter Mode Implementation for LEA Block Cipher.*
Si-Woo Eum, Min-Ho Song, Sang-Won Kim, Hwa-Jeong Seo
Journal of the Korea Institute of Information and Communication Engineering (JKIICE), 2025.
3. *Improved Parallel ARIA Optimization Implementation on 64-Bit ARMv8 Processor.*
Si-Woo Eum, Hwa-Jeong Seo
Journal of the Korea Institute of Information and Communication Engineering (JKIICE), 2024.
4. *Optimal Implementation of Lightweight Block Cipher PIPO on CUDA GPGPU.*
Hyun-Jun Kim, **Si-Woo Eum**, Hwa-Jeong Seo
Journal of the Korea Institute of Information Security & Cryptology (JKIISC), 2022.
5. *Implementation of LEA Lightweight Block Cipher GCM Operation Mode on 32-Bit RISC-V.*
Si-Woo Eum, Hyeok-Dong Kwon, Hyun-Ji Kim, Yu-Jin Yang, Hwa-Jeong Seo
Journal of the Korea Institute of Information Security & Cryptology (JKIISC), 2022.
6. *Optimized Implementation of PIPO Lightweight Block Cipher on 32-Bit RISC-V Processor.*
Si-Woo Eum, Kyung-Bae Jang, Gyeong-Ju Song, Min-Woo Lee, Hwa-Jeong Seo
KIPS Transactions on Computer and Communication Systems (KTCCS), 2022.
7. *Speed-Optimized Implementation of Fixslicing AES-CTR Using Pre-Computation on 32-Bit RISC-V.*
Si-Woo Eum, Hyun-Jun Kim, Min-Joo Sim, Gyeong-Ju Song, Hwa-Jeong Seo
Journal of the Korea Institute of Information Security & Cryptology (JKIISC), 2022.
8. *Optimized Parallel Implementation of PIPO Lightweight Block Cipher on 64-Bit ARM Processors.*
Si-Woo Eum, Hyeok-Dong Kwon, Hyun-Jun Kim, Kyung-Bae Jang, Hyun-Ji Kim, Jae-Hoon Park, Gyeong-Ju Song, Min-Joo Sim, Hwa-Jeong Seo
KIPS Transactions on Computer and Communication Systems (KTCCS), 2021.

Preprints

1. *Performance Analysis of a Thread Pool-Based Parallel Execution Model for Hybrid Post-Quantum TLS 1.3 Handshakes.*
Si-Woo Eum, Min-Ho Song, Hwa-Jeong Seo
Cryptology ePrint Archive, 2026.
2. *KpqClean Ver2: Comprehensive Benchmarking and Analysis of KpqC Algorithm Round 2 Submissions.*
Min-Joo Sim, **Si-Woo Eum**, Gyeong-Ju Song, Min-Woo Lee, Sang-Won Kim, Min-Ho Song, Hwa-Jeong Seo
Cryptology ePrint Archive, 2024.
3. *Analysis of Parallel Implementation of Pilsung Block Cipher on Graphics Processing Unit.*
Si-Woo Eum, Hyun-Jun Kim, Min-Ho Song, Hwa-Jeong Seo
Cryptology ePrint Archive, 2023.

Conference Presentations

International Conferences

1. *Maximizing ARIA-CTR Performance Through Kernel-Level Memory Optimization on the Apple M1 GPU.*
Si-Woo Eum, Min-Ho Song, Min-Joo Sim
Information Security and Cryptology – ICISC 2025: 28th International Conference, 2025.
2. *Implementation of SM4 Block Cipher on CUDA GPU and Its Analysis.*
Si-Woo Eum, Hyun-Jun Kim, Hyeok-Dong Kwon, Kyung-Bae Jang, Hyun-Ji Kim, Hwa-Jeong Seo
International Conference on Platform Technology and Service (PlatCon), 2022.

Domestic Conferences (Korea)

1. *Analysis of Multi-Party Threshold Cryptography Trends Based on NIST Threshold Call.*
CISC-S, 2026.

2. *Performance Analysis and Optimization of HCTR2 Mode Based on ARIA Block Cipher.*
CISC-W, 2025.
3. *Optimized GPU-based Camellia-CTR Implementation through Bank Conflict Minimization.*
CISC-S, 2025.
4. *Trends in Optimized Implementation of NIST PQC Algorithms.*
CISC-S, 2024.
5. *Performance Analysis of Re-encryption for Database Security Enhancement.*
CISC-W, 2023.
6. *Research Trends in Job Scheduling Techniques for Performance Improvement on Multi-GPU Systems.*
CISC Chungcheong Chapter Conference, 2023.
7. *GPU-Accelerated Implementation of Argon2.*
CISC-S, 2023.
8. *Trends in NTT Implementation on ARMv8.*
CISC Yeongnam Chapter Conference, 2022.
9. *Trends in FFT Implementation on ARMv8.*
CISC Chungcheong Chapter Conference, 2022.
10. *Optimized Implementation of SKINNY Tweakable Block Cipher on ARMv8.*
ASK, 2022.
11. *Implementation of SKINNY Tweakable Block Cipher on RISC-V.*
CISC-W, 2021.
12. *Optimized Implementation of LEA-CTR on 32-bit RISC-V.*
CISC Honam Chapter Conference, 2021.
13. *Optimized Parallel Implementation of Lightweight Block Cipher PIPO on 32-bit RISC-V.*
ACK, 2021.
14. *Implementation of Fixslicing AES-CTR Mode on RISC-V Using Pre-computation Tables.*
CISC Chungcheong Chapter Conference, 2021.
15. *Optimized Implementation of Lightweight Block Ciphers SIMON and SPECK on RISC-V Processors.*
CISC-S, 2021.
16. *Parallel Implementation of Block Cipher PIPO on GPU.*
CISC-S, 2021.
17. *Optimized Parallel Implementation of Block Cipher PIPO on 64-bit ARM Processors.*
ASK, 2021.

Projects

Role: Participating Researcher (all projects listed below).

- Jan. 1, 2026 – Dec. 31, 2027
Development of Core Technologies and Platforms based on Blockchain & Training of Future Innovative Talent (IITP)
- May 4, 2026 – Nov. 30, 2026
Analysis of Domestic and International Homomorphic Encryption Application Cases and Development of Usage Guidelines (Korea Cryptography Forum)
- Apr. 1, 2026 – Oct. 31, 2026
Research on Implementation Guidelines for KpqC Algorithms and Application Scenarios in Lightweight Environments (NSR)
- Jul. 1, 2025 – Dec. 31, 2027
Development of S/HBOM-based Supply Chain Protection and Security Enhancement Technology for Industrial Control Systems (IITP)
- Jul. 17, 2025 – Oct. 17, 2025
Implementation and Testing of KpqC Post-Quantum Cryptographic Algorithms (ETRI)
- May 1, 2025 – Nov. 30, 2025
Research on Advancement of National Network Security Framework (N²SF) and National Cryptography Application Criteria for PQC Transition (Korea Cryptography Forum)
- Apr. 1, 2025 – Dec. 31, 2028
Development and Demonstration of PQC-based Joint Certificate PKI Infrastructure Technology (IITP)
- May 1, 2025 – Nov. 30, 2025
Research on Implementation Optimization Technology for KpqC Competition Selected Algorithms (NSR)

- Jul. 1, 2024 – Nov. 30, 2024
Research on Satellite Cybersecurity Policy and Technology (Korea Cryptography Forum)
- Apr. 1, 2024 – Sep. 30, 2024
Research on Implementation and Practical Application Technology of Post-Quantum Cryptography (NSR)
- Apr. 1, 2023 – Oct. 31, 2023
Research on Performance Comparison and Analysis Technology through Implementation of KpqC Competition Algorithms (NSR)
- Feb. 1, 2023 – Nov. 1, 2023
Research on US NIST-PQC Key Exchange Techniques and Performance Comparison (Korea Cryptography Society)
- Apr. 1, 2021 – Dec. 31, 2023
Development of GPU/ASIC-based High-Speed Design and Implementation Technology for Cryptographic Algorithms (IITP)
- Jun. 1, 2020 – Feb. 28, 2023
Research on NIST Post-Quantum Cryptography Benchmarks on ARM Cortex-M4 and RISC-V (NRF)

Awards

- 2025 National Cryptography Contest: Excellence Award.
- 2024 National Cryptography Contest: Excellence Award.
- 2024 Side-Channel Information Analysis Competition: Chair's Award.
- 2023 Side-Channel Information Analysis Competition: Award.
- 2023 Hansung University Graduate School: Best Master's Thesis Award.
- 2022 Side-Channel Analysis Competition: KISA President's Award.
- 2022 National Cryptography Contest: Special Award.
- 2022 PlatCon'22: Best Paper Award.
- 2022 1st "Digital Transformation Convergence Ideas" Contest: Grand Prize.
- 2022 Quantum Information Competition: Special Award.
- 2022 CISC-S: Best Paper Award.
- 2021 AI+Security Idea Contest: Grand Prize.
- 2021 ACK: Best Paper Award.
- 2021 CISC-S: Best Paper Award.
- 2021 ASK: Best Paper Award.
- 2020 CISC Honam Chapter Conference: Best Paper Award.
- 2020 3rd Side-Channel Analysis Competition: Chair's Award.

Technical Skills

- **Programming Languages:** C, Python, Assembly (ARM, RISC-V)
- **Parallel & SIMD:** ARM NEON, CUDA, Metal (Apple Silicon)
- **Target Platforms:** ARM Cortex-M, ARMv8 (AArch64), RISC-V (RV32/RV64), NVIDIA GPU, Apple M1 GPU
- **Cryptography:** Post-Quantum Cryptography (KpqC, NIST PQC), Symmetric-Key Ciphers (AES, ARIA, LEA, PIPO, SM4, HIGHT, SEED), Hash Functions (LSH, Argon2), Hash-Based Signatures (XMSS, SPHINCS+)
- **Protocols & Frameworks:** TLS 1.3 (Hybrid PQC), PKI/X.509, KpqClean Benchmark Suite